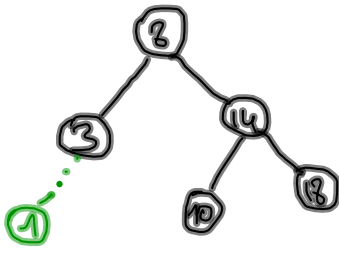
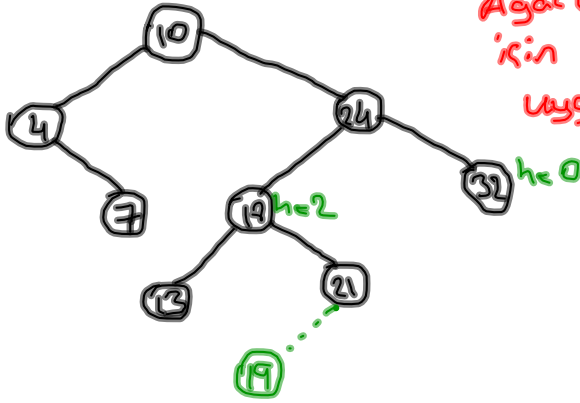




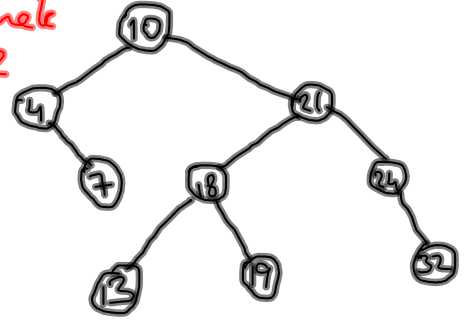
## AVL ağaclarında ekleme



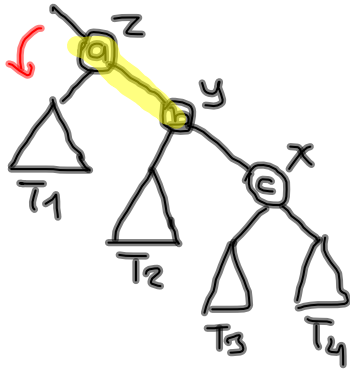
1 eklendiğinde yükseklik dengesi bozulmadı.



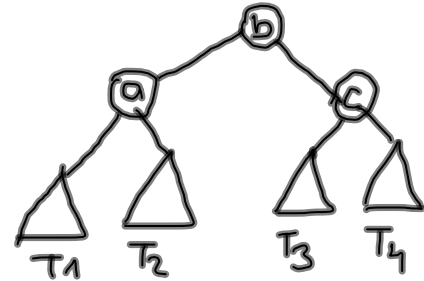
Ağacı dengelemek için döndürme uygularız



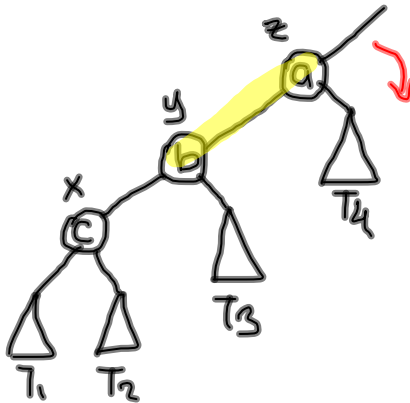
Bir AVL ağacında düğümleri döndürme:  
Tek döndürme (Single Rotation)



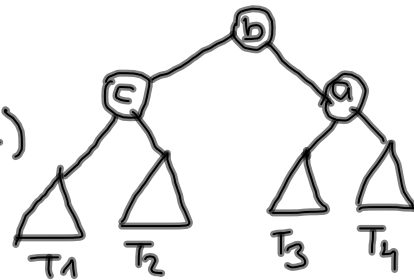
zig-zig  
Left\_Rotate(z)



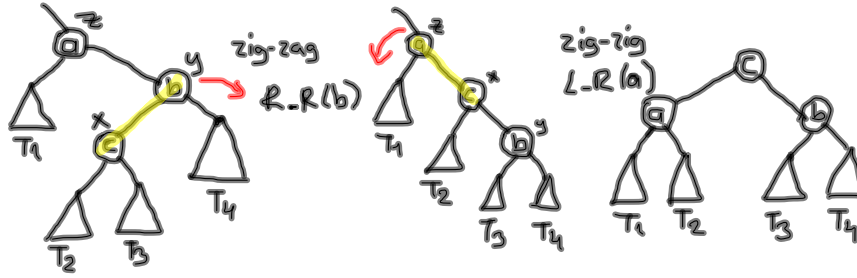
$$T_1 \leq z \leq T_2 \leq y \leq T_3 \leq x \leq T_4$$



zig-zig  
Right\_Rotate(z)

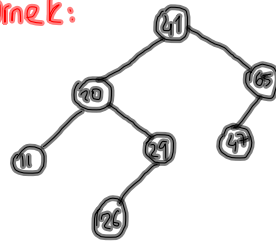


## Gifit Öndürme ( Double Rotation)

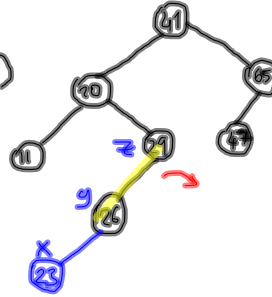
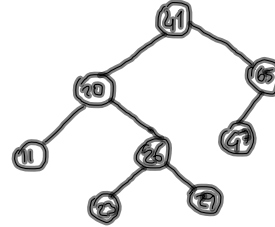


## Simetrisi

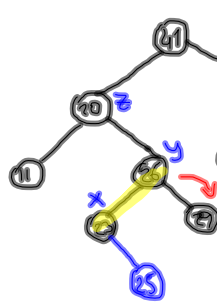
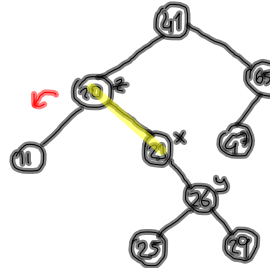
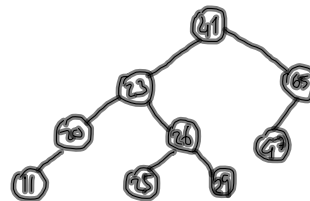
Örnek:



Insert(23)

zig-zig  
Right\_R(29)

insert(25)

zig-zig  
Right\_R(26)zig-zig  
Left\_R(20)

NOT: Eklediğimiz düğümden köke doğru olan yolda dengeyi bozan ilk düğüm z dir. z'nin o yol üzerindeki torunu x'tir.

